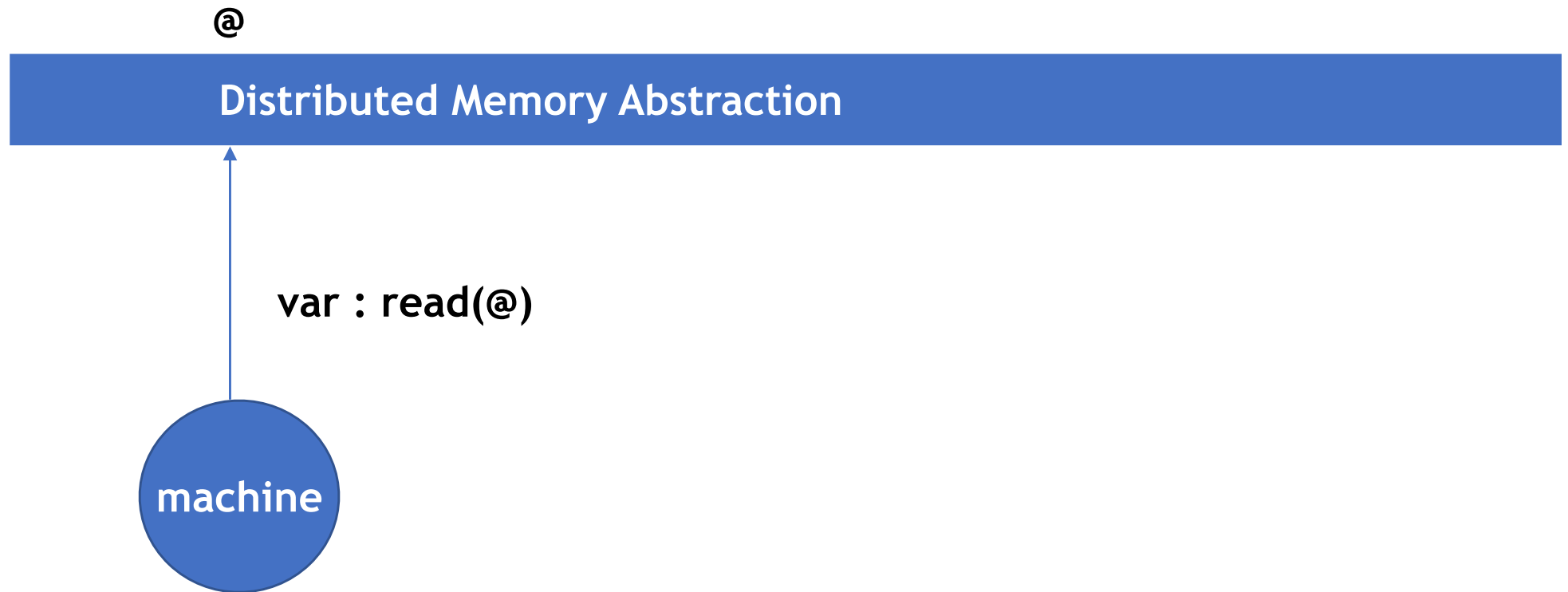


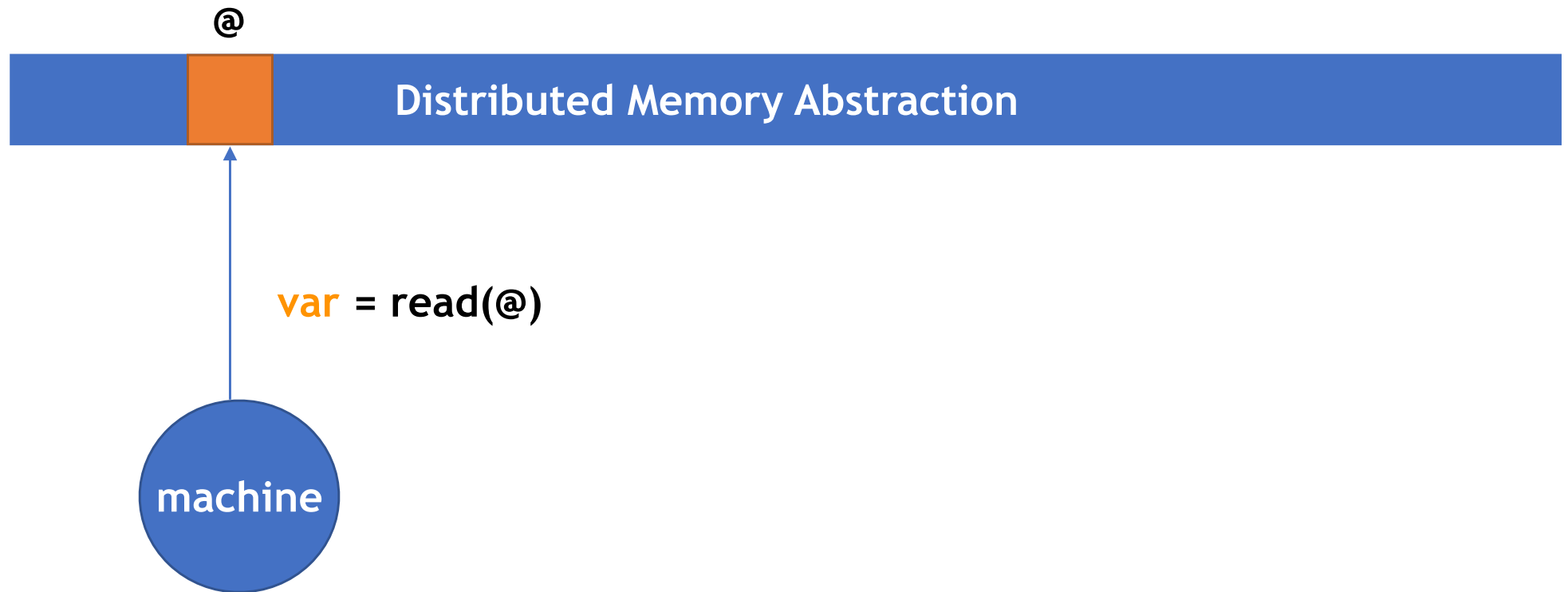
Project 3: DSM (Distributed Shared Memory)

- Distributed Shared Memory Abstraction



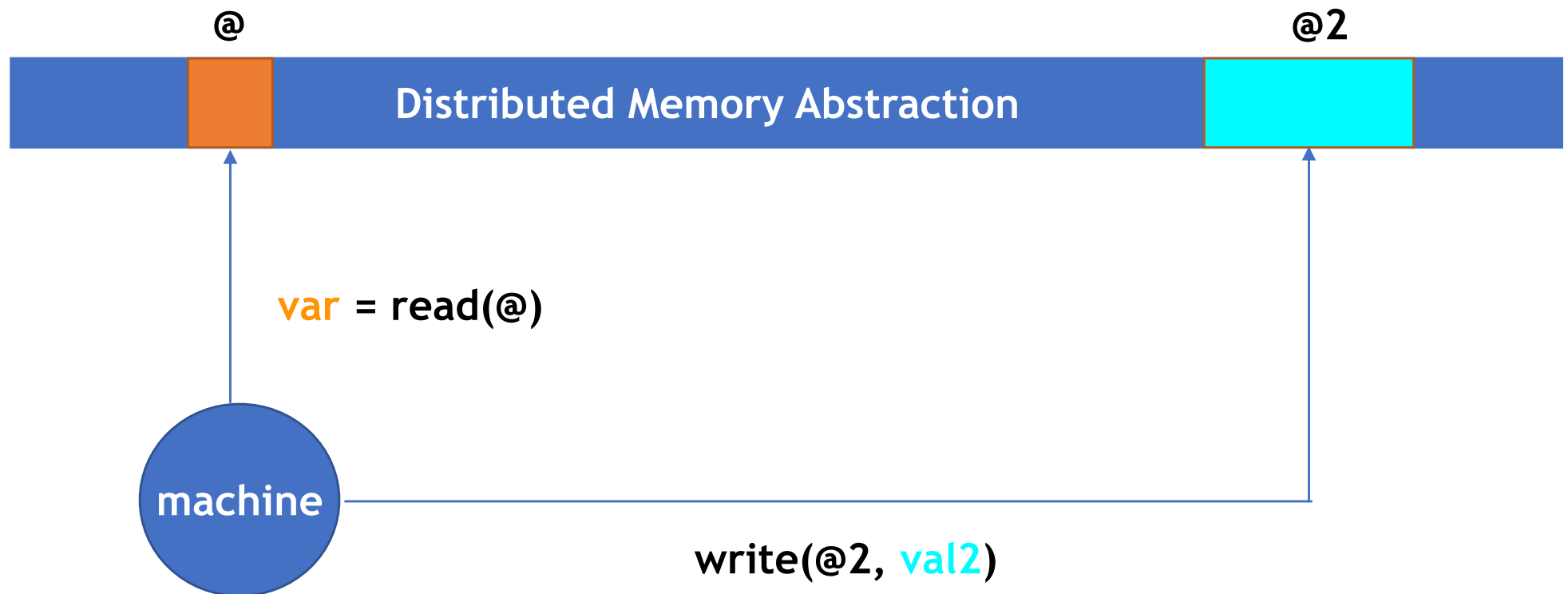
Project 3: DSM (Distributed Shared Memory)

- Distributed Memory Abstraction



Project 3: DSM (Distributed Shared Memory)

- Distributed Memory Abstraction



Project 3: DSM (Distributed Shared Memory)

Interface

- `value read(address)`
- `write(address, value)`

1. You need to decide **what is an address** and how to manage addresses
2. You need to decide **what kind of values** you will manipulate

- `value read(address, size)`
- `write(address, size, value)`

Project 3: DSM

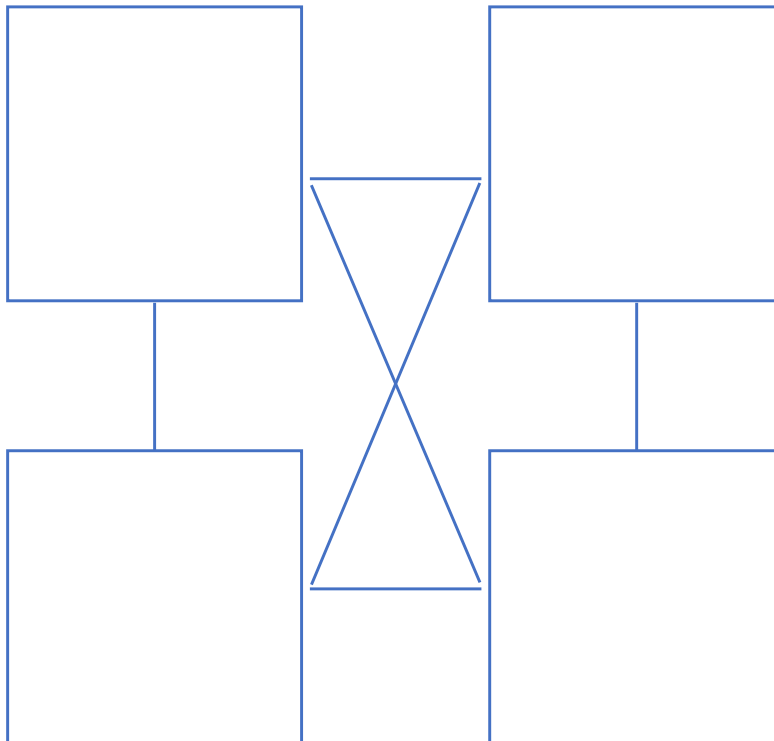
(Distributed Shared Memory)

- The memory is distributed over several physical machines
 - You need to decide how to distribute the memory
- The machines are interconnected into a distributed topology
 - You need to choose the topology
- You need to decide what is an address and how to manage addresses
- You need to decide what kind of values you will manipulate

Project 3: DSM

EXAMPLE Setting

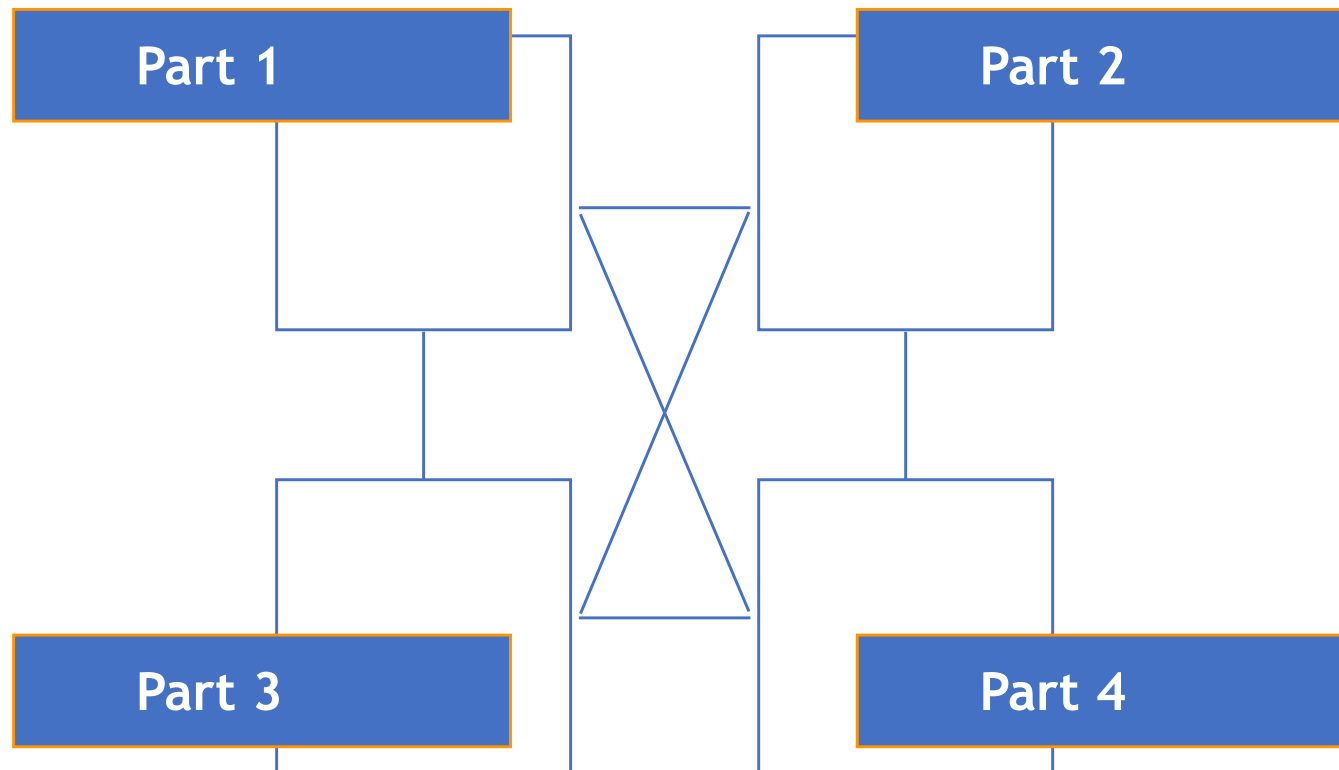
Distributed Shared Memory Abstraction



Physical machines in a grid

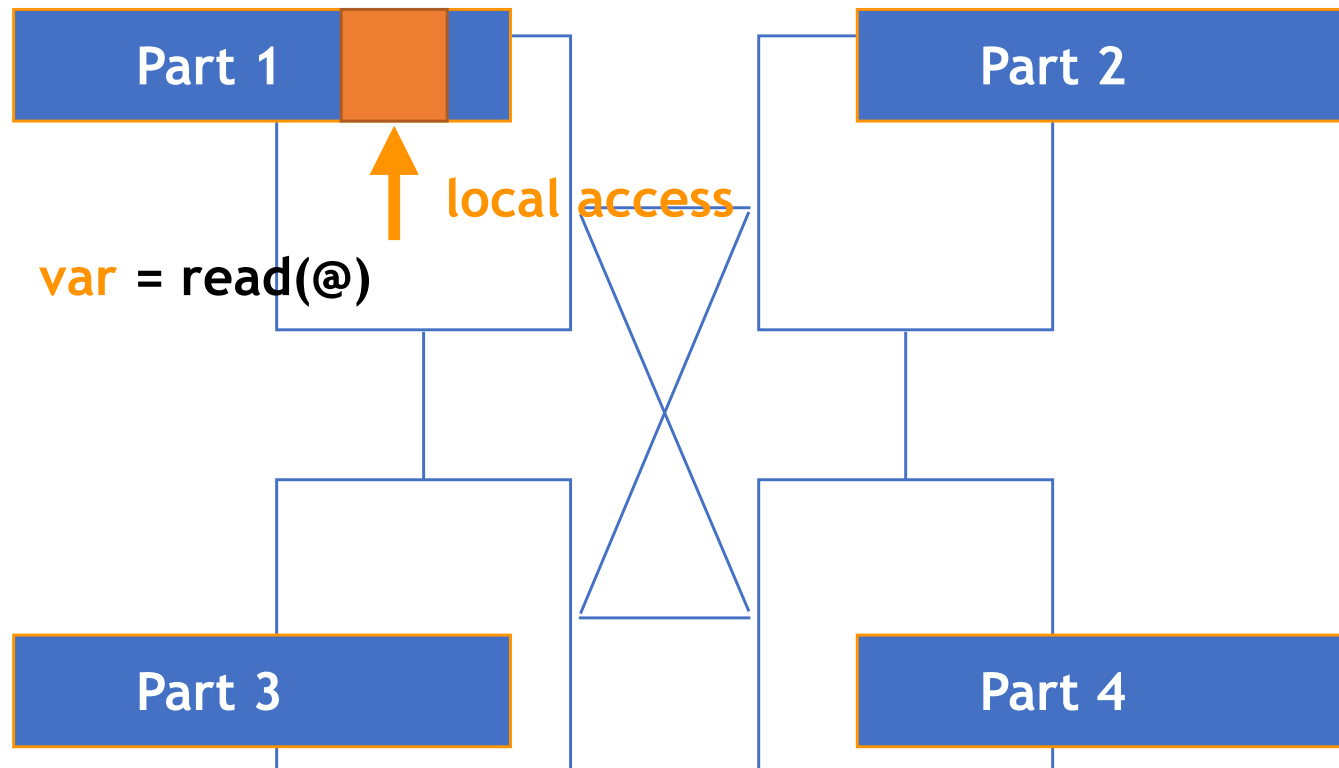
Project 3: DSM

EXAMPLE with Partitioning



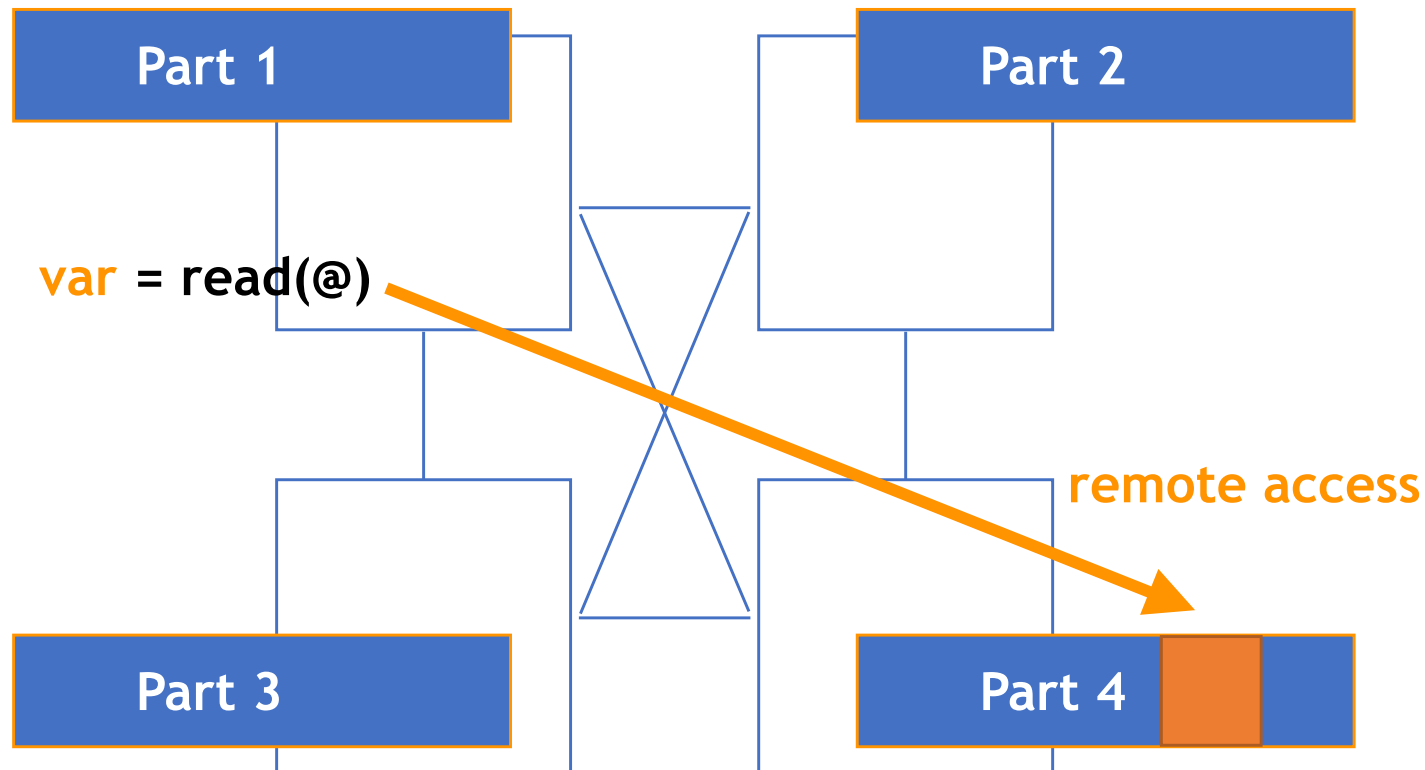
Project 3: DSM

EXAMPLE with Partitioning



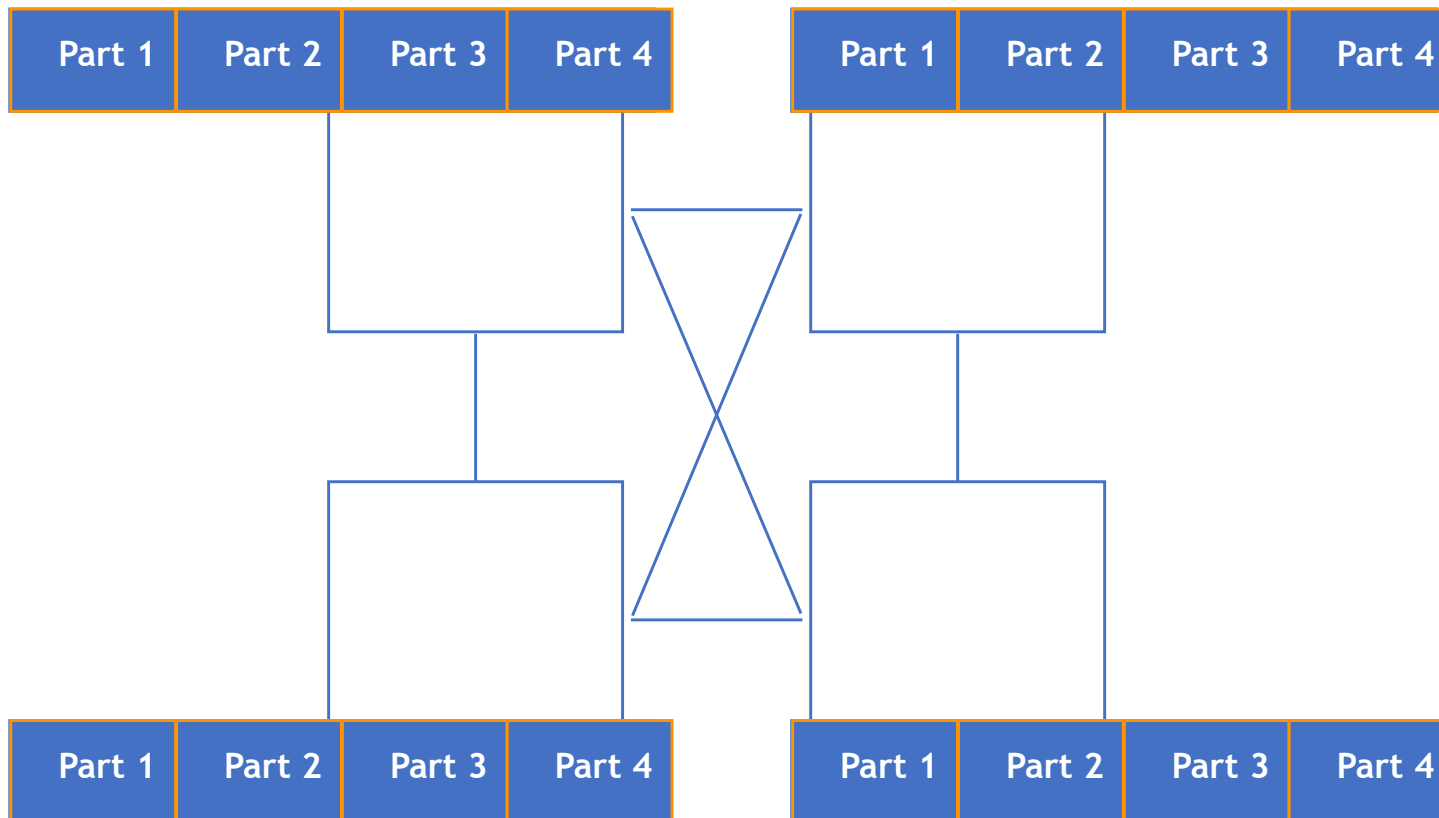
Project 3: DSM

EXAMPLE with Partitioning



Project 3: DSM

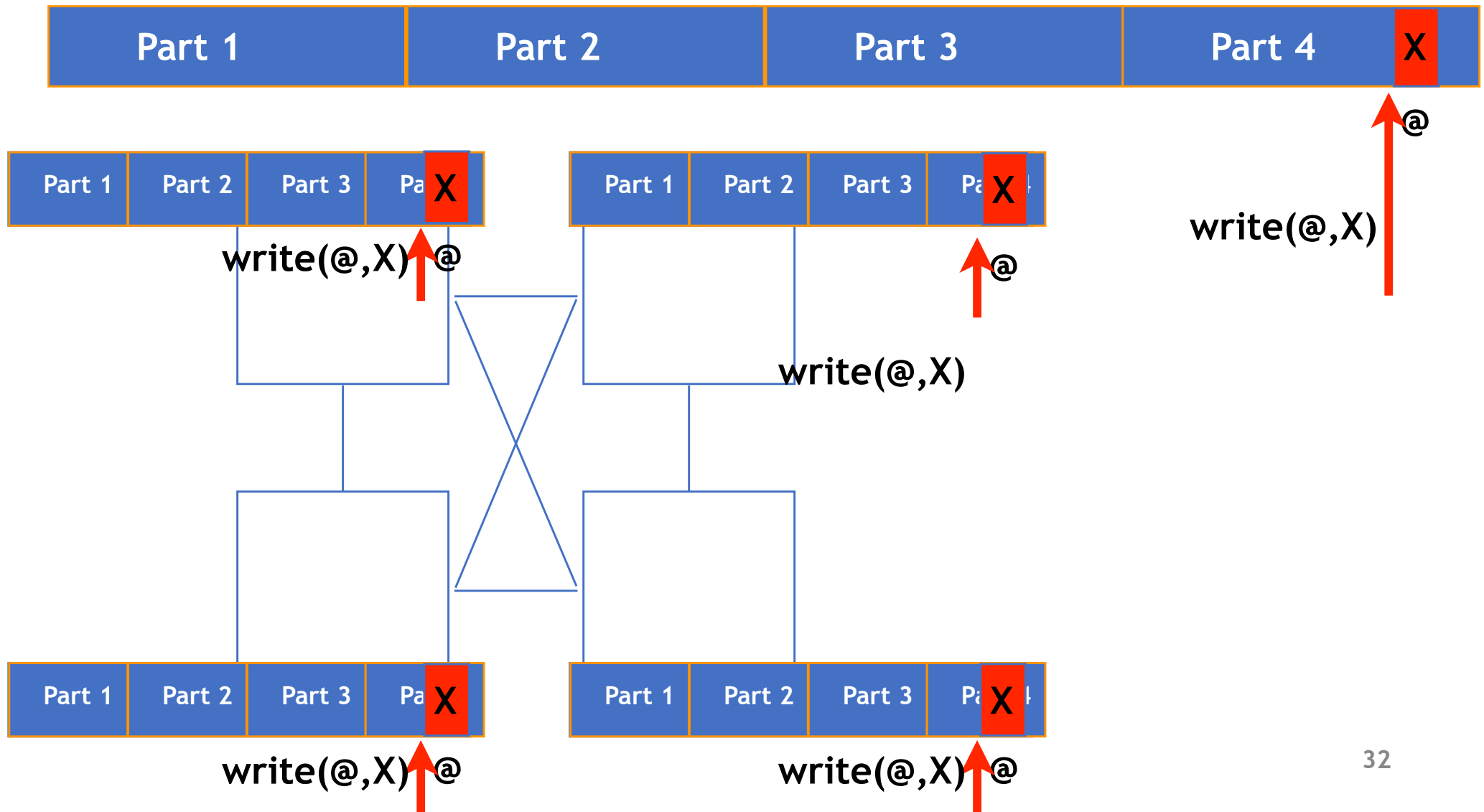
EXAMPLE with Replication



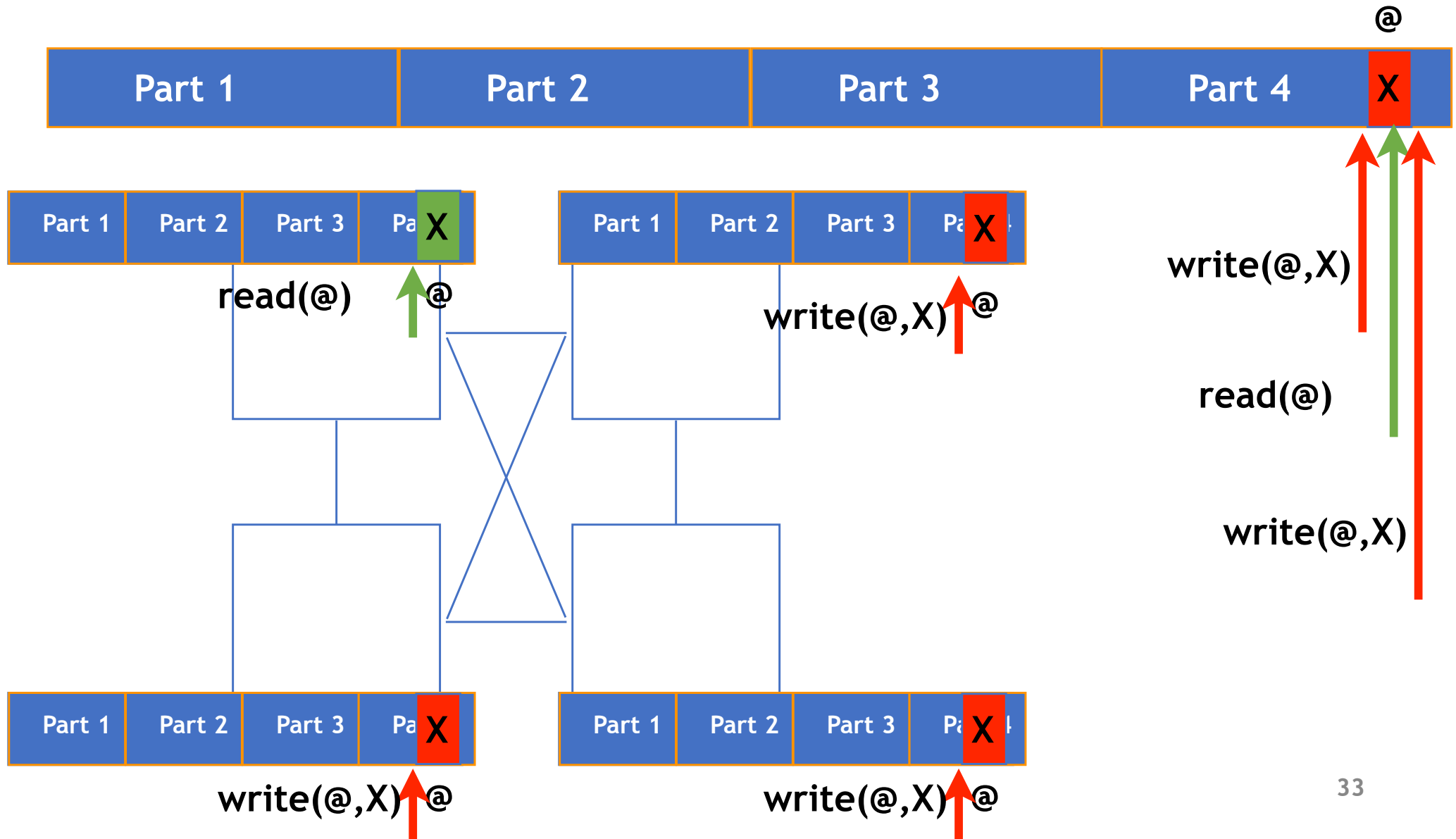
The big problem is **consistency** (this is really complicated)

- If there are only read operations, no problem
- The pb is when there are write operations
- What if two write operations concern the same address?
 - Which one will go first?
 - What value will the others observe?
(if they read from this address)

Strong consistency



Consistency?



References (on Moodle)

1. Fekete, A. D., Ramamritham, K.: Consistency Models for Replicated Data. In Charron- Bost, B., Pedone, F., Schiper, A., eds. : Replication LNCS 5959. Springer-Verlag, Berlin (2010) 1-17
2. Burckhardt, S.: Principles of Eventual Consistency. Foundations and Trends Programming Languages 1(1-2), 1-150 (2014)